Series & Binomial Theorem

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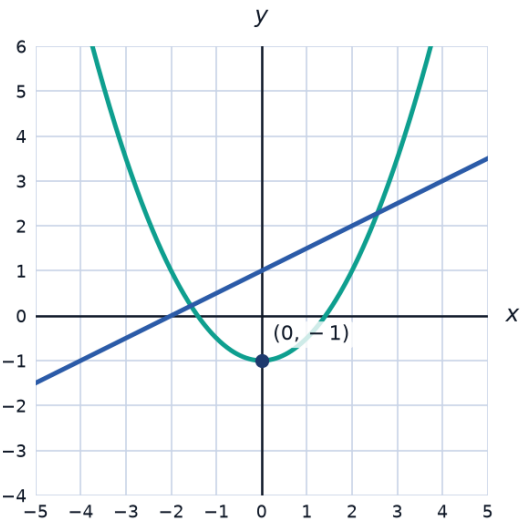
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Arithmetic Progressions (AP)**

A particle moves along a straight line. Its velocity, v m/s, at time t seconds is given by $v = 3t - 2$. Find the displacement of the particle during the first 4 seconds of its motion.

2 **[Foundational]** • **Target: Geometric Progressions (GP)**

The diagram shows the graph of the function $y = f(x)$. The graph is transformed by a sequence of operations. The resulting graph, $y = g(x)$, is also shown. The function $f(x)$ is given by $f(x) = ax^2 + bx + c$, where $a, b,$ and c are constants. The vertex of the graph of $y = f(x)$ is at $(2, -1)$, and it passes through the point $(4, 3)$. The graph of $y = g(x)$ is obtained by translating the graph of $y = f(x)$ by $(-1\ 3)$ followed by a reflection in the y-axis. Find the values of $a, b,$ and c.



3 **[Foundational]** • **Target: Sum to Infinity of a Convergent GP**

A committee of 3 people is to be selected from a group of 5 people. The number of ways to form this committee is given by the combination formula ${}^nC_r = \frac{n!}{r!\,(n-r)!}$. How many different committees can be formed?

4 **[Foundational]** • **Target: The Binomial Theorem for Positive Integers**

The diagram shows a right-angled triangle ABC, with the right angle at D. The altitude from D to the hypotenuse BC is drawn. Given that BD = 4 cm and DC = 9 cm, find the length of the altitude AD.

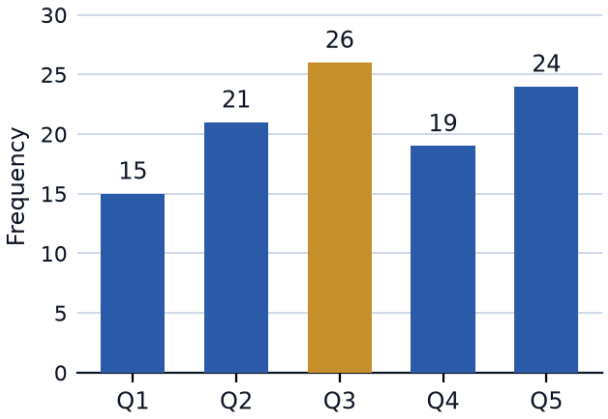
5 **[Foundational]** • **Target: Arithmetic Progressions (AP)**

A particle of mass 3 kg is at rest on a smooth horizontal surface. A horizontal force of magnitude P N is applied to the particle, causing it to accelerate at a rate of 4 m/s^2. Find the value of P.

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6 [Foundational] • Target: Geometric Progressions (GP)

The two-way table shows the results of a survey of 100 students on their preferred mode of transport to school and their year group. | | Car | Bus | Walk | Total | |--------------|-----|-----|------|-------| | Year 1 | 15 | 20 | 10 | 45 | | Year 2 | 25 | 18 | 12 | 55 | | Total | 40 | 38 | 22 | 100 | What is the probability that a randomly selected student is a Year 1 student who prefers to travel by car?



7 [Foundational] • Target: Sum to Infinity of a Convergent GP

A 3 kg particle on a rough surface, $\mu = 0.4$. Horizontal force F N. (a) Find the limiting friction. (b) Acceleration when $F = 20$ N. (c) Time to travel 5 m from rest.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: The Binomial Theorem for Positive Integers

A factory tests 30 independent components, each with a fixed probability of being defective; X is the number of defective components. Is X binomially distributed?

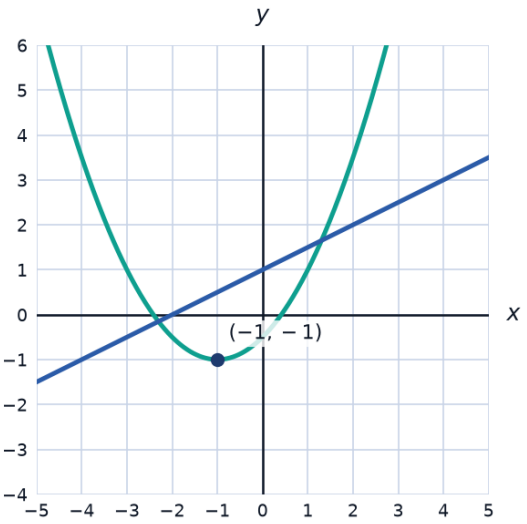
9 [Intermediate] • Target: Arithmetic Progressions (AP)

A survey asks 50 randomly and independently selected people the same yes/no question; X is the number who say yes. Is X binomially distributed?

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10 [Intermediate] • Target: Geometric Progressions (GP)

A machine has a useful power output of 883 W and wastes 367 W as heat. Find its efficiency as a percentage, correct to 1 decimal place.



11 [Intermediate] • Target: Sum to Infinity of a Convergent GP

Solve $\cos(x) = -\frac{\sqrt{2}}{2}$ for $0^\circ \le x \le 360^\circ$.

12 [Intermediate] • Target: The Binomial Theorem for Positive Integers

Solve $\cos(x) = -\frac{1}{2}$ for $0^\circ \le x \le 360^\circ$.

13 [Intermediate] • Target: Arithmetic Progressions (AP)

Solve $\tan(x) = -\frac{\sqrt{3}}{3}$ for $0^\circ \le x \le 360^\circ$.

14 [Intermediate] • Target: Geometric Progressions (GP)

Solve $\tan(x) = \frac{\sqrt{3}}{3}$ for $0^\circ \le x \le 360^\circ$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

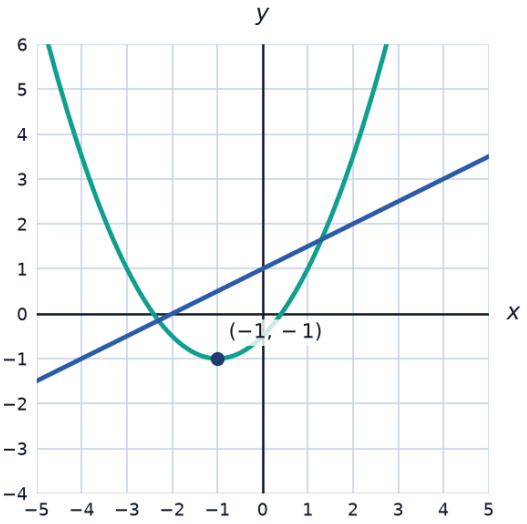
15 [Challenging] • Target: Sum to Infinity of a Convergent GP • Cross-Lesson: 7.01 Statistics & Probability

A particle of mass 5 kg slides down a rough plane inclined at 30° to the horizontal, experiencing a constant friction force of 0 N. Taking $g = 9.8$ m/s^2, find its acceleration down the slope.

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16 [Challenging] • Target: The Binomial Theorem for Positive Integers • Cross-Lesson: 8.01 Functions: Composite & Inverse

Evaluate $\int_1^2 (6x^2 - 5x + 1)\,dx$.



17 [Challenging] • Target: Arithmetic Progressions (AP) • Cross-Lesson: 9.01 Circular Measure

Evaluate $\int_{-2}^4 (3x^2 - 3x - 1)\,dx$.

18 [Challenging] • Target: Geometric Progressions (GP) • Cross-Lesson: 1.01 Indices, Surds & Quadratics

Given $X \sim N(\mu, \sigma^2)$, $P(X < 99) = 0.8$ and $P(X < 83) = 0.2$, find μ, correct to the nearest integer.

19 [Challenging] • Target: Sum to Infinity of a Convergent GP • Cross-Lesson: 11.01 Kinematics

Evaluate $\int_1^3 (6x^2 + 2x + 3)\,dx$.

20 [Challenging] • Target: The Binomial Theorem for Positive Integers • Cross-Lesson: 12.01 Forces & Equilibrium

A particle of mass m kg is placed on a rough inclined plane. The plane is inclined at an angle α to the horizontal, where $\tan\alpha = \frac{3}{4}$. The particle is on the point of sliding down the plane. The coefficient of static friction between the particle and the plane is μ_s. Given that the acceleration due to gravity is $g = 9.8$ m/s^2, find the value of μ_s.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Arithmetic Progressions (AP)

An arithmetic progression has first term $a = 7$ and common difference $d = 4$. Find the 20th term u_{20}.

a. 83

b. 87

c. 79

d. 80

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22 [Foundational] • Target: Arithmetic Progressions (AP)

Find the sum of the first 25 terms of the AP: $3, 8, 13, 18, \ldots$

a. 1575

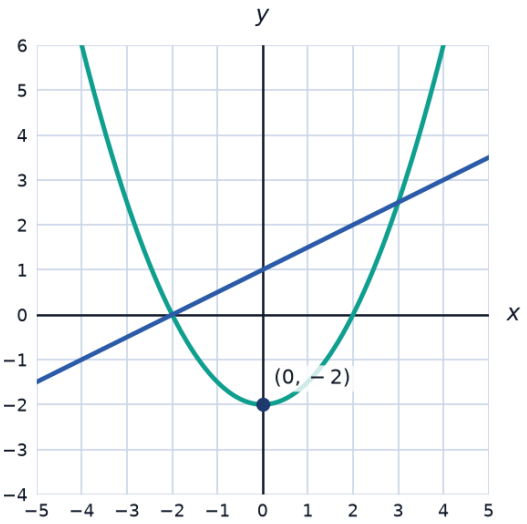
b. 1625

c. 1500

d. 1550

23 [Foundational] • Target: Arithmetic Progressions (AP)

In an AP, the 5th term is 18 and the 12th term is 39. Find the first term a and common difference d.



a. $a = 6, d = 3$

b. $a = 3, d = 3$

c. $a = 9, d = 3$

d. $a = 6, d = 4$

24 [Foundational] • Target: Geometric Progressions (GP)

A geometric progression has first term $a = 5$ and common ratio $r = 2$. Find the 8th term u_8.

a. 640

b. 1280

c. 320

d. 512

25 [Foundational] • Target: Geometric Progressions (GP)

Find the sum of the first 6 terms of the GP: $3, -6, 12, -24, \ldots$

a. -63

b. 63

c. -127

d. -65

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26 [Foundational] • Target: Geometric Progressions (GP)

The 2nd term of a GP is 12 and the 5th term is 324. Find the common ratio r.

a. 3

b. 2

c. 4

d. $\sqrt{3}$

27 [Foundational] • Target: Sum to Infinity of a Convergent GP

State the condition for the sum to infinity of a geometric progression to exist.

a. $|r| < 1$

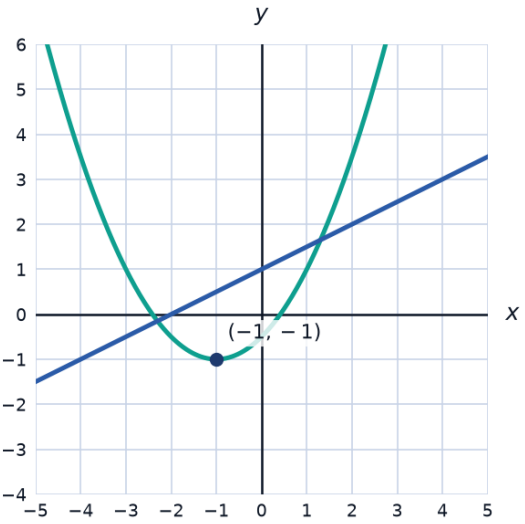
b. $r > 1$

c. $r \leq 1$

d. $|r| > 1$

28 [Foundational] • Target: Sum to Infinity of a Convergent GP

Find the sum to infinity of the GP: $16, 12, 9, 6.75, \ldots$



a. 64

b. 48

c. 32

d. 72

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Sum to Infinity of a Convergent GP

Express the recurring decimal $0.\overline{27} = 0.272727\ldots$ as an exact fraction in lowest terms using an infinite geometric series.

a. $\frac{3}{11}$

b. $\frac{27}{100}$

c. $\frac{9}{33}$

d. $\frac{7}{25}$

30 [Intermediate] • Target: The Binomial Theorem for Positive Integers

Find the coefficient of x^3 in the expansion of $(2 + 3x)^5$.

a. 1080

b. 720

c. 540

d. 1440

31 [Intermediate] • Target: The Binomial Theorem for Positive Integers

Find the term independent of x in the expansion of $\left(x^2 + \frac{2}{x}\right)^6$.

a. 240

b. 60

c. 160

d. 120

32 [Intermediate] • Target: The Binomial Theorem for Positive Integers

In the expansion of $(1 + ax)^6$, the coefficient of x^2 is 60. Given $a > 0$, find a.

a. 2

b. 4

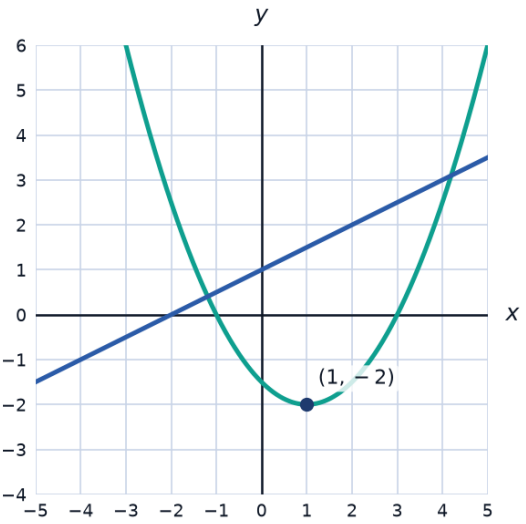
c. 3

d. 5

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33 [Intermediate] • Target: Arithmetic Progressions (AP)

A geometric series has first term $a = -10$ and common ratio $r = 1/2$. Find S_6, the sum of the first 6 terms.



a. -315/16

b. -155/8

c. -475/16

d. -635/32

34 [Intermediate] • Target: Geometric Progressions (GP)

A geometric series has first term $a = 13$ and common ratio $r = 5/2$. Find S_5, the sum of the first 5 terms.

a. 67431/32

b. 13403/16

c. 2639/8

d. 13611/16

35 [Intermediate] • Target: Sum to Infinity of a Convergent GP

A geometric series has first term $a = -3$ and common ratio $r = 2/3$. Find S_5, the sum of the first 5 terms.

a. -211/27

b. -292/27

c. -65/9

d. -665/81

36 [Intermediate] • Target: The Binomial Theorem for Positive Integers

A geometric series has first term $a = 6$ and common ratio $r = -4/3$. Find S_5, the sum of the first 5 terms.

a. -50/9

b. -962/81

c. 524/27

d. 362/27

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37 [Intermediate] • Target: Arithmetic Progressions (AP)

Find the sum of the first 40 terms of an arithmetic sequence with first term -11 and common difference -2.

a. -2000

b. -2004

c. -1994

d. -2003

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Geometric Progressions (GP) • Cross-Lesson: 1.01 Indices, Surds & Quadratics

A geometric series has first term $a = 6$ and common ratio $r = 3/2$. Find S_6, the sum of the first 6 terms.

a. 1995/16

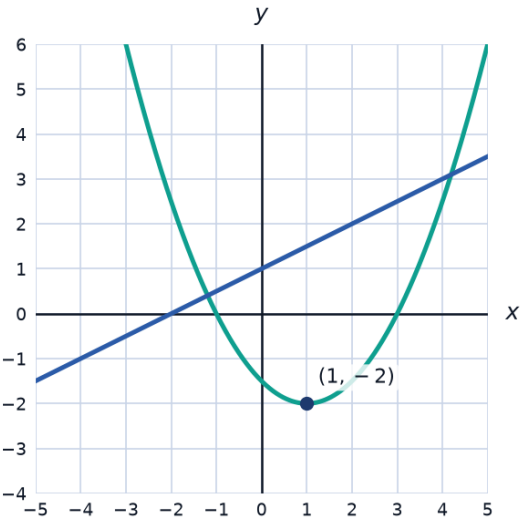
b. 2091/16

c. 633/8

d. 6177/32

39 [Challenging] • Target: Sum to Infinity of a Convergent GP • Cross-Lesson: 11.01 Kinematics

A geometric series has first term $a = 2$ and common ratio $r = 5/2$. Find S_6, the sum of the first 6 terms.



a. 5187/16

b. 5219/16

c. 25999/32

d. 1031/8

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40 [Challenging] • Target: The Binomial Theorem for Positive Integers • Cross-Lesson: 12.01 Forces & Equilibrium

A geometric series has first term $a = 1$ and common ratio $r = 4/3$. Find S_6, the sum of the first 6 terms.

a. 3367/243

b. 14197/729

c. 3610/243

d. 781/81

41 [Challenging] • Target: Arithmetic Progressions (AP) • Cross-Lesson: 13.01 Newton's Laws

Find the sum of the first 40 terms of an arithmetic sequence with first term -9 and common difference -2.

a. -1920

b. -1926

c. -1916

d. -1923

42 [Challenging] • Target: Geometric Progressions (GP) • Cross-Lesson: 14.01 Energy, Work & Power

Find the sum of the first 46 terms of an arithmetic sequence with first term -9 and common difference -7.

a. -7662

b. -7659

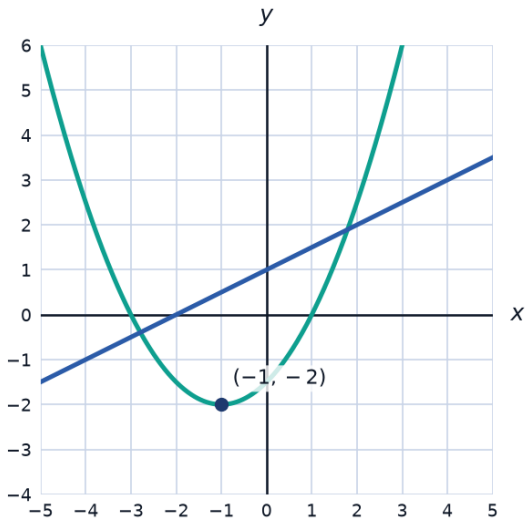
c. -7668

d. -7669

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43 [Challenging] • Target: Sum to Infinity of a Convergent GP • Cross-Lesson: 15.01 Representation of Data

Find the sum of the first 17 terms of an arithmetic sequence with first term -12 and common difference -3.



a. -612

b. -608

c. -607

d. -613

44 [Challenging] • Target: The Binomial Theorem for Positive Integers • Cross-Lesson: 16.01 Permutations & Combinations

A geometric series has first term $a = 3$ and common ratio $r = 1/2$. Find S_5, the sum of the first 5 terms.

a. 93/16

b. 141/16

c. 45/8

d. 189/32

45 [Challenging] • Target: Arithmetic Progressions (AP) • Cross-Lesson: 17.01 Discrete Random Variables

Find the sum of the first 32 terms of an arithmetic sequence with first term -12 and common difference 7.

a. 3074

b. 3077

c. 3088

d. 3078

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